Exercise sheet: Week 8 - solutions

1. Recall from Week 7 Question 1 exercise,
   $\underline{X}$ is a random sample from $\mathrm{Poi}(\lambda)$, we choose as prior $\lambda \sim \mathrm{Gamma}(\alpha, \beta)$. Then, given a random sample $\underline{x}$ we have

$$\begin{aligned} \pi(\lambda|\underline{x}) &\propto \pi(\lambda) L(\lambda; \underline{x}) \\ &\propto \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \frac{1}{\Pi\, x_i!} \lambda^{\sum x_i} e^{-n\lambda}, \quad \text{where } n \text{ is the sample size,} \\ &\propto \lambda^{\alpha+\sum x_i - 1} e^{-(\beta+n)\lambda}, \end{aligned} \tag{1}$$

   implying the posterior distribution is $\lambda|\underline{x} \sim \mathrm{Gamma}(\alpha + \sum x_i,\ \beta + n)$.

   (a) Given $n = 10$, $\sum_{i=1}^{10} x_i = 28$, and prior $\lambda \sim \mathrm{Gamma}(0.5, 0.5)$,

$$\lambda \mid \underline{x} \sim \mathrm{Gamma}(\alpha + \textstyle\sum x_i,\ \beta + n) = \mathrm{Gamma}(28.5,\ 10.5).$$

   Hence,

$$\text{posterior mean} = \frac{\alpha + \sum x_i}{\beta + n} = \frac{28.5}{10.5} = \frac{19}{7} \approx 2.714,$$

   and

$$\text{posterior variance} = \frac{\alpha + \sum x_i}{(\beta + n)^2} = \frac{28.5}{10.5^2} = \frac{38}{147} \approx 0.2585.$$

   (b) To get the equal-tailed 95% credible interval, note that

$$2 \times 10.5 \times \lambda \sim \mathrm{Gamma}(28.5, 0.5) \sim \chi^2_{57}$$

   or

$$21\lambda \sim \mathrm{Gamma}(28.5, 0.5) \sim \chi^2_{57}$$

   One can use tables to get the 97.5% and 2.5% points of χ^2_{57} (via linear interpolation between the percentage points of χ^2_{50} and χ^2_{60}).
   Otherwise, use R to calculate `qchisq(0.025,57), qchisq(0.975,57)`

$$\chi^2_{0.025,\,57} \approx 38.02, \qquad \chi^2_{0.975,\,57} \approx 79.75.$$

   Therefore, the required 95% credible interval is

$$\left(\frac{\chi^2_{0.025,\,57}}{21}, \frac{\chi^2_{0.975,\,57}}{21}\right) = \left(\frac{38.02}{21}, \frac{79.75}{21}\right) \approx (1.81,\ 3.79).$$

(c) With $\alpha' = 28.5$ and $\beta' = 10.5$, the posterior-predictive mass function for a new observation $Y \sim \text{Poisson}(\lambda)$ is

$$\begin{aligned} f_Y(y \mid \underline{x}) = P(Y = y \mid \underline{x}) &= \int_0^\infty P(Y = y \mid \lambda)\, \pi(\lambda \mid \underline{x})\, d\lambda \\ &= \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!} \cdot \frac{\beta'^{\alpha'}}{\Gamma(\alpha')} \lambda^{\alpha'-1} e^{-\beta'\lambda}\, d\lambda \\ &= \frac{\beta'^{\alpha'}}{\Gamma(\alpha')\, y!} \int_0^\infty \lambda^{\alpha'+y-1} e^{-(\beta'+1)\lambda}\, d\lambda \\ &= \frac{\beta'^{\alpha'}}{\Gamma(\alpha')\, y!} \cdot \frac{\Gamma(\alpha'+y)}{(\beta'+1)^{\alpha'+y}} \\ &= \frac{\Gamma(\alpha'+y)}{\Gamma(\alpha')\, y!} \left(\frac{\beta'}{\beta'+1}\right)^{\alpha'} \left(\frac{1}{\beta'+1}\right)^{y}. \end{aligned}$$

Hence, substituting $\alpha' = 28.5$ and $\beta' = 10.5$,

$$f_Y\left(y \,\middle|\, \sum_{i=1}^{10} x_i = 28\right) = \frac{\Gamma(28.5+y)}{\Gamma(28.5)\, y!} \left(\frac{10.5}{11.5}\right)^{28.5} \left(\frac{1}{11.5}\right)^{y}, \quad y = 0, 1, 2, \ldots$$

Therefore, $P(Y = y \mid \sum_{i=1}^{10} x_i = 28)$ follows a Negative Binomial (Poisson–Gamma mixture) predictive distribution with parameters $r = \alpha' = 28.5$, and $p = \frac{\beta'}{\beta'+1} = \frac{10.5}{11.5} \approx 0.913$.

Remark. The Gamma integral identity used above is

$$\int_0^\infty \lambda^{k-1} e^{-c\lambda}\, d\lambda = \frac{\Gamma(k)}{c^k}, \qquad (k > 0,\ c > 0),$$

applied with $k = \alpha' + y$ and $c = \beta' + 1$.

2. For a random sample $x_1, \ldots, x_n$ from $\text{Poisson}(\lambda)$ the likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} \prod_{i=1}^{n} \frac{1}{x_i!}.$$

The log-likelihood is

$$\ell(\lambda) = \log L(\lambda) = \Big(\sum_{i=1}^{n} x_i\Big) \log \lambda - n\lambda - \sum_{i=1}^{n} \log(x_i!).$$

Differentiating with respect to λ gives

$$\ell'(\lambda) = \frac{\sum_{i=1}^n x_i}{\lambda} - n, \qquad \ell''(\lambda) = -\frac{\sum_{i=1}^n x_i}{\lambda^2}.$$

The Fisher information is the negative expectation of the second derivative:

$$I(\lambda) = -\mathbb{E}[\ell''(\lambda)] = -\mathbb{E}\left[-\frac{\sum_{i=1}^n X_i}{\lambda^2}\right] = \frac{\mathbb{E}\big[\sum_{i=1}^n X_i\big]}{\lambda^2} = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$$

Jeffreys' prior is proportional to the square root of the Fisher information:

$$\pi(\lambda) \propto \sqrt{I(\lambda)} \propto \sqrt{\frac{n}{\lambda}} \propto \lambda^{-1/2}, \qquad \lambda > 0,$$

since the factor $\sqrt{n}$ is constant in λ. Hence

$$\pi(\lambda) \propto \lambda^{-1/2}, \quad \lambda > 0.$$

3. Assume $X_1, \ldots, X_n$ are *i.i.d.* $\mathrm{Bin}(m, p)$. The likelihood is

$$L(p) = \prod_{i=1}^n \binom{m}{x_i} p^{x_i}(1-p)^{m-x_i} \propto p^{\sum_{i=1}^n x_i}(1-p)^{nm-\sum_{i=1}^n x_i}.$$

The log-likelihood is

$$\ell(p) = \log L(p) = \Big(\sum_{i=1}^n x_i\Big)\log p + \Big(nm - \sum_{i=1}^n x_i\Big)\log(1-p).$$

Differentiating $\ell(p)$ twice with respect to p gives

$$\ell'(p) = \frac{\sum_{i=1}^n x_i}{p} - \frac{nm - \sum_{i=1}^n x_i}{1-p}, \qquad \ell''(p) = -\frac{\sum_{i=1}^n x_i}{p^2} - \frac{nm - \sum_{i=1}^n x_i}{(1-p)^2}.$$

The Fisher information is the negative expectation of the second derivative:

$$\begin{aligned} I(p) &= -\mathbb{E}[\ell''(p)] \\ &= \frac{\mathbb{E}\big[\sum_{i=1}^n X_i\big]}{p^2} + \frac{\mathbb{E}\big[nm - \sum_{i=1}^n X_i\big]}{(1-p)^2} \\ &= \frac{nmp}{p^2} + \frac{nm(1-p)}{(1-p)^2} = \frac{nm}{p(1-p)} \end{aligned}$$

Jeffreys' prior is proportional to the square root of the Fisher information:

$$\pi(p) \propto \sqrt{I(p)} = \sqrt{\frac{nm}{p(1-p)}} \propto p^{-1/2}(1-p)^{-1/2}, \qquad 0 < p < 1.$$

Recognizing this as a Beta distribution,

$$\pi(p) \propto p^{\frac{1}{2}-1}(1-p)^{\frac{1}{2}-1} \sim \mathrm{Beta}\big(\tfrac{1}{2}, \tfrac{1}{2}\big)\,.$$

Remark. The Jeffreys prior does not depend on n or m; even for Binomial observations with known m, the prior is still $\mathrm{Beta}(0.5, 0.5)$.

4. Assume $X_1, \ldots, X_n$ are *i.i.d.* Geometric(p), where

$$\mathbb{P}(X_i = x_i) = p(1-p)^{x_i - 1}, \qquad x_i = 1, 2, 3, \ldots$$

The likelihood is

$$L(p) = \prod_{i=1}^{n} p(1-p)^{x_i-1} = p^n (1-p)^{\sum_{i=1}^n (x_i - 1)} = p^n (1-p)^{\sum_{i=1}^n x_i - n}.$$

The log-likelihood is

$$\ell(p) = \log L(p) = n \log p + \Big(\sum_{i=1}^{n} x_i - n \Big) \log(1-p).$$

Differentiating twice with respect to p gives

$$\ell'(p) = \frac{n}{p} - \frac{\sum_{i=1}^n X_i - n}{1-p}, \qquad \ell''(p) = -\frac{n}{p^2} - \frac{\sum_{i=1}^n X_i - n}{(1-p)^2}.$$

The Fisher information is the negative expectation of the second derivative:

$$\begin{aligned} I(p) &= -\mathbb{E}[\ell''(p)] \\ &= \frac{n}{p^2} + \frac{\mathbb{E}\big[\sum_{i=1}^n X_i - n\big]}{(1-p)^2} \\ &= \frac{n}{p^2} + \frac{n\,(\mathbb{E}[X_1] - 1)}{(1-p)^2} \\ &= \frac{n}{p^2} + \frac{n(\frac{1}{p} - 1)}{(1-p)^2} \quad (\because\ \mathbb{E}[X_1] = \frac{1}{p}) \\ &= \frac{n}{p^2} + \frac{n(\frac{1-p}{p})}{(1-p)^2} = \frac{n}{p^2} + \frac{n}{p(1-p)} = \frac{n}{p^2(1-p)}. \end{aligned}$$

Jeffreys' prior is proportional to the square root of the Fisher information:

$$\pi(p) \propto \sqrt{I(p)} = \sqrt{\frac{n}{p^2(1-p)}} \propto p^{-1}(1-p)^{-1/2}, \qquad 0 < p < 1.$$

Comment: The Jeffreys prior for the Geometric(p) model is

$$\pi(p) \propto p^{-1}(1-p)^{-1/2}, \qquad 0 < p < 1,$$

while for the Binomial(m, p) model it is

$$\pi(p) \propto p^{-1/2}(1-p)^{-1/2}.$$

Compared with the Binomial case, as $p \to 0$, the Geometric prior places more weight, reflecting the stronger influence of small p values on the likelihood in the Geometric distribution. As $p \to 1$, both priors behave similarly. This shows Jeffreys' prior adapts to the information in the model.

5. (a) We have $T \sim \text{Exp}(\lambda)$ and $\lambda \sim \Gamma(1, 2)$. Take a random sample of size $n = 5$ and observe $\sum t_i = 3$. Therefore,

$$(\lambda|\underline{t}) \sim \Gamma(1+n, 2+\sum t_i) \sim \text{Gamma}(6, 5).$$

(b) Let Z denote the lifetime of the next component. We require a posterior predictive density of Z,

$$\begin{aligned}
\pi(z|\underline{t}) &= \int_0^\infty \pi(\lambda|\underline{t})\,\pi(z|\lambda)d\lambda \\
&= \frac{5^6}{\Gamma(6)} \int_0^\infty \lambda^5 e^{-5\lambda} \lambda e^{-\lambda Z} d\lambda \\
&= \frac{5^6}{\Gamma(6)} \int_0^\infty \lambda^6 e^{-\lambda(5+Z)} d\lambda \\
&= \frac{5^6}{5!} \frac{\Gamma(7)}{(5+z)^7} \\
&= \frac{6 \cdot 5^6}{(5+z)^7}
\end{aligned}$$

Thus,

$$\mathbf{P}(Z > z \,|\, \underline{t}) = \int_z^\infty \frac{6 \cdot 5^6}{(5+z')^7}\, dz' = \left(\frac{5}{5+z}\right)^6$$

and

$$\mathbf{P}(Z < 0.25 \,|\, \underline{t}) = 1 - \left(\frac{5}{5.25}\right)^6 = 0.254 \quad \text{to 3 d.p.}$$